

IFSA Karate Mumbai — Website Master Reference Guide

Indian Fit Sports Academy | ifsaacademy.in

Last updated: May 2026 | Covers all phases of the website build

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1. Website Overview

The IFSA website is a full-stack Node.js website with a dark-mode design (navy + amber/gold). It has:

- **7 public pages** + 1 admin panel + 2 system pages (404, offline)
- **Dynamic content** — photos, schedule, instructors, pricing, etc. are all managed from the admin panel with no code changes needed
- **PWA support** — can be installed on phones like an app
- **WhatsApp integration** — announcements can be broadcast via WhatsApp gateway

Page	File	Purpose
Home	<code>INDEX.html</code>	Main landing page
About	<code>about.html</code>	Academy history, affiliations, team
Gallery	<code>gallery.html</code>	Masonry photo/video gallery
Schedule	<code>calender.html</code>	Events calendar + timetable
Documents	<code>documents.html</code>	Downloadable PDFs
Pricing	<code>pricing.html</code>	Plans + booking form
Admin	<code>admin.html</code>	Password-protected CMS
404	<code>404.html</code>	Custom error page
Offline	<code>offline.html</code>	PWA offline fallback

2. How the Site Works (Architecture)

```

Browser (visitor)
  |
  ▼
NGINX (reverse proxy, HTTPS, port 80/443)
  |
  ▼
Node.js / Express ↔ JSON files in /data/
(server.js, port 3000) ↔ /uploads/ folder
  |
  ▼
Static HTML files (INDEX.html, about.html, etc.)

```

- The **server** (`server.js`) runs on Node.js and handles all API calls and file uploads
- **All content data** is stored in JSON files inside the `/data/` folder — no database needed
- **Uploaded files** (photos, videos, PDFs) go into `/uploads/`
- The **admin panel** (`admin.html`) talks to the server via API calls using a JWT token
- The **service worker** (`sw.js`) caches the site for offline use

Tech Stack

Layer	Technology
Server	Node.js + Express
Auth	bcrypt + JWT (8-hour sessions)
File uploads	Multer
Image processing	Sharp (auto-compress to WebP)
Excel export	ExcelJS
Frontend CSS	Tailwind CSS (CDN) + custom STYLE.css
PWA	Service Worker + manifest.json

3. File & Folder Map

```
project-root/
|
├─ INDEX.html           ← Homepage
├─ about.html
├─ gallery.html
├─ calender.html
├─ pricing.html
├─ documents.html
├─ grading.html
├─ payment.html
├─ admin.html           ← Admin panel (password protected)
├─ 404.html
├─ offline.html
|
├─ server.js            ← Main backend server (Node.js)
├─ change-password.js   ← Run on server to change admin password
├─ STYLE.css            ← Global custom styles + CSS variables
├─ theme.js             ← (Planned) Shared dark/light theme manager
├─ cookie-consent.js    ← (Planned) Cookie consent banner
|
├─ sw.js               ← Service worker (PWA caching)
├─ manifest.json       ← PWA manifest
```

```
|— sitemap.xml           ← Static sitemap for Google
|— robots.txt           ← Crawler rules
|
|— data/                 ← All content stored as JSON
|   |— locations.json
|   |— instructors.json
|   |— testimonials.json
|   |— timetable.json
|   |— hero-video.json
|   |— slideshow-order.json
|   |— stats.json        ← (Planned) Editable hero stat numbers
|   |— achievements.json
|   |— timetable-slots.json ← (Planned) Custom slot names
|   |— admin-config.json ← (Planned) Named multi-admin support
|
|— uploads/              ← All uploaded files live here
|   |— slideshow/
|   |— gallery/
|   |— documents/
|   |— hero/
|   |— instructors/
|   |— testimonials/
|   |— achievements/     ← (Planned) Achievement images
|
|— gallery-data.json
|— document-data.json
|— schedule-data.json
|— announcement-data.json
|— admin-log.json
|— bookings-data.json
|— bookings.xlsx         ← Auto-updated Excel export of bookings
```

4. Admin Panel — How to Operate

Logging In

1. Open your browser and go to: `ifsaacademy.in/admin.html`
2. Type your admin password and click **Login**
3. You have **8 hours** before the session expires (timer shown top-right)
4. After **5 wrong attempts**, you are locked out for 15 minutes

⚠ The admin panel URL is not linked on the public site — keep it private.

Section-by-Section Guide

Hero Video

- Upload an MP4/WebM (max 100MB) to show a full-screen video on the homepage
- If no video is uploaded, the photo slideshow plays instead
- Click **Remove Hero Video** to go back to the slideshow

1 — Home Page Slideshow

- Upload JPG/PNG/WebP photos for the scrolling hero background
- **Drag and drop** thumbnails to reorder → click **Save Order**
- Click the trash icon to delete a photo

2 — Gallery

- Upload photos (JPG/PNG/WebP) or videos (MP4/WebM) — max 50MB each
- Always fill in Month, Year, and Alt Text before uploading
- Tick checkboxes + click **Delete Selected** to bulk delete

3 — Documents

- Upload PDF files with a clear title (e.g. "Tournament Registration Form 2025")
- These appear on the public Documents page as downloadable files

4 — Calendar / Schedule

- Add events with: Date, Title, and Type
 - **General** = regular update
 - **Exam** = belt grading
 - **Tournament** = competition
 - **Holiday** = no class days
- Events are colour-coded and sorted automatically by date

5 — Notice Board (Announcements)

- Post a notice with optional expiry date — it disappears automatically on that date
- Leave expiry blank to keep it up permanently

- Use this for: class cancellations, new batches, fee changes

6 — Locations & Pricing

- Add each branch with: Name, Subtitle, Google Maps embed URL, Beginner plan, Advanced plan
- **To edit pricing:** delete the branch and re-add with new prices (no edit button yet)
- Google Maps embed URL: go to Google Maps → Share → Embed a map → copy only the URL inside `src="..."`

7 — Instructors

- Add with: Name, Role, Rank, Front photo (portrait), Back photo (action shot), Bio
- Cards appear on the Homepage and About page with a 3D flip effect

8 — Testimonials

- Add student quotes with: Name, Belt Rank, Quote, Photo (optional)
- These rotate automatically on the homepage every 5 seconds

9 — Timetable

- Add class slots: Day + Time Slot + Batch Name + Display Time + Instructor
- If a slot already exists for that Day + Time, it is replaced
- Delete individual slots from the list below the form

10 — Trial Bookings

- Every form submission from the website appears here
- Update status: **Pending** (yellow) → **Contacted** (blue) → **Enrolled** (green)
- Call Pending leads within 24 hours for best conversion

11 — Activity Log

- Automatic record of all admin actions: what changed, when, and from which IP
- Click **Refresh** to see latest entries
- Read-only — nothing to do here

Things Currently Hardcoded (Require Developer to Change)

These cannot be changed from the admin panel yet — a developer must edit `INDEX.html`

directly:

- Hero stats bar (500+ Students, 15+ Years, 50+ Medals, 4 Locations)
- Programs section text (Kids / Adult / Competition)
- Footer contact details (phone, email, address)
- "Why Choose IFSA" section text
- Belt progression timeline descriptions

Note: Stats will become editable once Fix 5 (see Section 10) is implemented.

5. Common Tasks — Quick Reference

Task	Steps
Add class photos	Admin → Section 2 (Gallery) → Select files → Set month/year/alt text → Upload
Change class fees	Admin → Section 6 → Delete branch → Re-add with new prices
Post a notice	Admin → Section 5 → Type message → Set expiry date → Add
Add grading event	Admin → Section 4 → Date + Title + Type "Exam" → Add
Mark booking as contacted	Admin → Section 10 → Click status badge → Change to Contacted
Upload a PDF document	Admin → Section 3 → Select PDF → Type title → Upload
Update timetable	Admin → Section 9 → Pick Day + Slot → Enter details → Save Slot
Add an instructor	Admin → Section 7 → Fill all fields + upload 2 photos → Add Instructor
Change hero video	Admin → Hero Video section → Upload MP4 (max 100MB)
Remove hero video	Admin → Hero Video section → Click Remove Hero Video

6. API Endpoints Reference

All API routes are on the same server as the website. Admin-only routes require a Bearer token in the `Authorization` header.

Public Routes (No Auth Needed)

Method	Endpoint	What it does
GET	<code>/api/slideshow/images</code>	Returns slideshow photos
GET	<code>/api/hero-video</code>	Returns current hero video filename
GET	<code>/api/gallery/images</code>	Returns gallery photos + videos
GET	<code>/api/documents/list</code>	Returns downloadable PDFs
GET	<code>/api/locations</code>	Returns all branches + pricing
GET	<code>/api/instructors</code>	Returns instructor profiles
GET	<code>/api/schedule/list</code>	Returns calendar events
GET	<code>/api/announcement/list</code>	Returns active announcements
GET	<code>/api/timetable</code>	Returns weekly class grid
GET	<code>/api/testimonials</code>	Returns student testimonials
GET	<code>/api/achievements</code>	Returns achievements list
GET	<code>/api/stats</code>	Returns hero stat numbers (<i>planned</i>)
GET	<code>/api/timetable-slots</code>	Returns slot names (<i>planned</i>)
POST	<code>/api/bookings/add</code>	Submits a trial class booking
POST	<code>/api/admin/login</code>	Admin login, returns JWT
POST	<code>/api/admin/verify</code>	Verifies existing JWT

Admin-Only Routes (JWT Required)

Method	Endpoint	What it does
POST	<code>/api/slideshow</code>	Upload slideshow image
DELETE	<code>/api/slideshow/:filename</code>	Delete a slideshow image
PATCH	<code>/api/slideshow/reorder</code>	Save new image order

Method	Endpoint	What it does
POST	<code>/api/hero-video</code>	Upload hero video
DELETE	<code>/api/hero-video</code>	Remove hero video
POST	<code>/api/gallery</code>	Upload gallery image/video
DELETE	<code>/api/gallery/:filename</code>	Delete gallery item
POST	<code>/api/documents</code>	Upload PDF document
DELETE	<code>/api/documents/:filename</code>	Delete document
POST	<code>/api/locations</code>	Add a branch location
DELETE	<code>/api/locations/:id</code>	Delete a location
POST	<code>/api/instructors</code>	Add instructor
DELETE	<code>/api/instructors/:id</code>	Delete instructor
POST	<code>/api/schedule</code>	Add calendar event
DELETE	<code>/api/schedule/:id</code>	Delete event
POST	<code>/api/announcements</code>	Post announcement
DELETE	<code>/api/announcements/:id</code>	Delete announcement
POST	<code>/api/timetable</code>	Add/update timetable slot
DELETE	<code>/api/timetable/:id</code>	Delete timetable slot
POST	<code>/api/testimonials</code>	Add testimonial
DELETE	<code>/api/testimonials/:id</code>	Delete testimonial
POST	<code>/api/achievements</code>	Add achievement
PUT	<code>/api/achievements/:id</code>	Edit achievement
DELETE	<code>/api/achievements/:id</code>	Delete achievement
GET	<code>/api/bookings</code>	View all bookings
PATCH	<code>/api/bookings/:id</code>	Update booking status

Method	Endpoint	What it does
DELETE	<code>/api/bookings/:id</code>	Delete booking
GET	<code>/api/bookings/export</code>	Download bookings as Excel (<i>planned</i>)
GET	<code>/api/admin/log</code>	View activity log
POST	<code>/api/stats</code>	Update hero stats (<i>planned</i>)
POST	<code>/api/timetable-slots</code>	Add a custom slot (<i>planned</i>)
DELETE	<code>/api/timetable-slots/:index</code>	Delete a slot (<i>planned</i>)
POST	<code>/api/wa-contacts/import</code>	Import contacts CSV (<i>planned</i>)

7. Data Files Reference

All data is stored in plain JSON files. If you ever need to manually fix or inspect data, here is what each file contains:

File	Contents
gallery-data.json	Gallery image metadata (filename, month, year, alt text)
document-data.json	Document metadata (filename, title, date)
schedule-data.json	Calendar events (date, title, type)
announcement-data.json	Announcements (text, optional expiry date)
admin-log.json	Last 200 admin actions
bookings-data.json	Trial class booking form submissions
bookings.xlsx	Same bookings as Excel file
data/locations.json	Branch locations + pricing plans
data/instructors.json	Instructor profiles
data/testimonials.json	Student testimonials
data/timetable.json	Weekly class grid
data/hero-video.json	Current hero video filename
data/slideshow-order.json	Custom slideshow order
data/achievements.json	Achievements/medals list
data/stats.json	Hero stats numbers (<i>planned</i>)
data/timetable-slots.json	Custom slot names (<i>planned</i>)
data/admin-config.json	Named admin accounts (<i>planned</i>)
data/wa-config.json	WhatsApp gateway config
data/wa-contacts.json	WhatsApp broadcast contacts

Achievement JSON Structure

json

```
{
  "id": "ach_1",
  "icon": "🏆",
  "image": "/uploads/achievements/winner.jpg",
  "title": "National Championship Gold",
  "subtitle": "WKF Senior Kata – New Delhi",
  "year": "2024",
  "category": "tournament"
}
```

8. Testing the Website

Quick Health Check (Do This After Any Deployment or Change)

Open the website and check each of these:

- ☐ Homepage loads within 3 seconds
- ☐ Preloader animation plays and fades out
- ☐ Hero slideshow or video plays
- ☐ Stats bar numbers count up when scrolled to
- ☐ Locations section shows correct branches and maps
- ☐ Instructors section loads with flip cards
- ☐ Timetable grid is populated
- ☐ Testimonials carousel auto-advances
- ☐ Trial booking form submits (use test data, then delete from admin)
- ☐ Gallery page loads photos
- ☐ Schedule page shows upcoming events
- ☐ Documents page lists PDFs
- ☐ Pricing page shows correct prices
- ☐ WhatsApp FAB (floating button) visible bottom-right on every page
- ☐ Theme toggle works (dark ↔ light)
- ☐ On mobile: hamburger menu opens and closes

Testing the Admin Panel

1. Go to `/admin.html` and log in
2. Check each section loads without errors
3. Add a test item in each section → verify it appears on the public site
4. Delete the test item → verify it disappears

Testing the API Directly (for Developers)

Open your browser console or use a tool like Postman/curl:

```
bash

# Check if server is running
curl https://ifsaacademy.in/api/locations

# Test a booking submission
curl -X POST https://ifsaacademy.in/api/bookings/add \
  -H "Content-Type: application/json" \
  -d '{"name":"Test User","phone":"9999999999","age":"25","session":"morning"}'

# Check admin login (replace YOUR_PASSWORD)
curl -X POST https://ifsaacademy.in/api/admin/login \
  -H "Content-Type: application/json" \
  -d '{"password":"YOUR_PASSWORD"}'
```

Checking Server Logs (on the server)

```
bash

# If running with PM2
pm2 logs ifsa

# If running directly
node server.js

# Check if port 3000 is in use
lsof -i :3000
```

Performance Testing

Run a Lighthouse audit in Chrome:

1. Open Chrome DevTools (F12)
 2. Go to the **Lighthouse** tab
 3. Click **Analyze page load**
 4. Target scores: Performance > 80, SEO > 90, Accessibility > 80
-

9. Bug Fixes & Known Issues

The following bugs are confirmed and fixes have been designed. Status: **pending implementation**.

Bug 1 — Achievements Wall: No Animation (INDEX.html)

Problem: Achievements render as a static grid with no animation or image support.

Fix: Replace `#achievements-grid` with a Pinterest-style 3-card carousel:

- 3 portrait cards visible: center card scaled up (scale 1.05, full brightness), side cards dimmed (scale 0.92, brightness 0.55)
- Auto-advances every 5 seconds, pauses on hover
- Swipe support on mobile (touchstart/touchend)
- Dot indicators + Prev/Next arrows

Files to change: `INDEX.html` (carousel HTML + JS + CSS)

CSS to add:

CSS

```
.ach-card-wrap { transition: transform 0.45s ease, filter 0.45s ease, opacity 0.45s ease; }
.ach-card-center { transform: scale(1.05); filter: brightness(1); z-index: 2; }
.ach-card-side { transform: scale(0.92); filter: brightness(0.55); opacity: 0.85; }
```

Bug 2 — Testimonials: One Card Appears Multiple Times

Problem: `renderTestimonials()` renders ALL testimonial cards every time a slide happens. The carousel track width is not recalculated, causing the visible card to appear duplicated.

Fix in `INDEX.html`:

js

```
// WRONG (current):
track.innerHTML = tData.map((t, i) => `...card html...`).join('');

// CORRECT (fix):
// Render ONLY tData[tIdx] as a single centered card.
// Arrow buttons call testimonialSlide() → re-render single card.
track.innerHTML = buildCard(tData[tIdx]);
```

Files to change: INDEX.html

Bug 3 — Achievements Not Editable in Admin (admin.html)

Problem: The Add Achievement form HTML exists in admin.html but all JavaScript functions are missing — nothing works.

Fix: Add these missing JS functions to admin.html:

```
js

async function loadAchievements() {
  // GET /api/achievements → populate #ach-list
  // Each row: icon | title | year | Edit | Delete buttons
}

async function addAchievement() {
  // POST /api/achievements with form values as multipart/form-data
}

async function deleteAchievement(id) {
  // Confirm dialog → DELETE /api/achievements/:id
}

async function editAchievement(id) {
  // Fill form with existing data → change button to "Update"
  // On submit → PUT /api/achievements/:id
}
```

Also call loadAchievements() on page load.

Files to change: admin.html

Bug 4 — WA Contacts Panel Not Wired (admin.html)

Problem: WhatsApp contacts panel HTML exists but all JS is missing.

Fix: Add to `admin.html`:

```
js

async function loadWaContacts() { /* GET /api/wa-contacts */ }
async function addWaContact() { /* POST /api/wa-contacts */ }
async function deleteWaContact(id) { /* DELETE /api/wa-contacts/:id */ }
async function toggleWaContact(id) { /* PATCH /api/wa-contacts/:id/toggle */ }
```

Files to change: `admin.html`

Bug 5 — Google Reviews Admin Panel Not Wired (admin.html)

Problem: Google Reviews settings panel HTML exists but all JS is missing.

Fix: Add to `admin.html`:

```
js

async function loadGoogleReviewsAdmin() { /* GET settings */ }
async function saveGoogleReviewsSettings() { /* POST settings */ }
async function refreshGoogleReviews() { /* trigger refresh */ }
```

Files to change: `admin.html`

Bug 6 — Duplicate Belt Grading Section (INDEX.html)

Problem: Two belt sections exist: `#belt-grading-teaser` (has the "View Full Syllabus" button) and `#belt-progression`. The first section is a duplicate and should be removed.

Fix:

1. Delete the entire `<section id="belt-grading-teaser">` block from `INDEX.html`
2. Add the "View Full Syllabus" button inside `#belt-progression`, after the "Minimum 5-6 years" line:

```
html
```



```
<div class="text-center mt-8">
  <a href="grading.html" class="btn-shimmer inline-flex items-center gap-2
    bg-gradient-to-r from-brand-accent to-amber-600 text-black px-8 py-3
    rounded-xl font-bold text-lg hover:scale-105 transition transform
    shadow-[0_0_20px_rgba(251,191,36,0.35)]">
    View Full Syllabus →
  </a>
</div>
```

Files to change: `INDEX.html`

10. Planned Fixes & New Features

These are designed and ready to implement.

Fix A — Editable Hero Stats Bar

Goal: Allow admin to change "500+ Students", "15+ Years", etc. without touching code.

New file: `data/stats.json`

```
json
{
  "years": { "value": 15, "suffix": "+", "label": "Years" },
  "students": { "value": 500, "suffix": "+", "label": "Students" },
  "blackBelts": { "value": 50, "suffix": "+", "label": "Black Belts" },
  "locations": { "value": 3, "suffix": "", "label": "Locations" }
}
```

Server changes: Add `GET /api/stats` (public) and `POST /api/stats` (admin).

INDEX.html change: On load, fetch `/api/stats` and apply values to counter elements dynamically before the animation runs.

Admin panel change: Add a new "📊 Hero Stats" panel with 4 number inputs and a Save button.

Files to change: `server.js`, `INDEX.html`, `admin.html`

Fix B — Editable Timetable Slots

Goal: Allow admin to add/remove time slots (e.g. "Afternoon 3–5 PM") without code changes.

New file: `data/timetable-slots.json`

```
json

["Morning (6-8 AM)", "Evening (5-7 PM)"]
```

Server changes: Add `GET /api/timetable-slots`, `POST /api/timetable-slots`, `DELETE /api/timetable-slots/:index`.

Admin panel change: Add "Manage Slots" sub-panel inside the Timetable section, above the form.

Files to change: `server.js`, `admin.html`

Fix C — Named Multi-Admin Login with Activity Tracking

Goal: Different admins have their own passwords and names. The activity log shows who did what.

New file: `data/admin-config.json`

```
json

{
  "admins": [
    { "passwordHash": "<bcrypt hash>", "name": "Sensei Saurabh", "role": "Head Admin" },
    { "passwordHash": "<bcrypt hash>", "name": "Coordinator", "role": "Staff" }
  ]
}
```

JWT payload change:

```
js

// Current:  { role: 'admin' }
// New:      { role: 'admin', name: 'Sensei Saurabh' }
```

Activity log change: Each log entry adds a `who` field.

Admin panel change: Activity log shows  `WHO` column in purple.

Files to change: `server.js`, `admin.html`

Fix D — WhatsApp Broadcast Group Support

Goal: When posting an announcement, also send to a WhatsApp broadcast group (not just individual contacts).

wa-config.json change:

```
json
{
  "broadcastGroupId": "120363XXXXXXXXXX@broadcast",
  "broadcastEnabled": true
}
```

Server change: After the personal message loop, send one API call with the group JID.

CSV import: Admin can upload a CSV file of phone numbers to bulk-import contacts.

```
name,phone
Rahul Shah,919876543210
Priya Nair,919004383448
```

Files to change: `server.js`, `admin.html`

Fix E — Cookie Consent Banner

Goal: Legal GDPR compliance before running Google Ads or Analytics.

New file: `cookie-consent.js`

- On first visit: shows a bottom bar with "Accept All" / "Essential Only" / "Learn More"
- Saves choice to `localStorage` as `ifsa_cookie_consent`
- Google Analytics only loads after "Accept All" is clicked
- Never shown again after a choice is made

New page: `privacy.html` — lists all cookies and their purpose

Every HTML page change: Add one line in `<head>`:

```
html
```

```
<script src="/cookie-consent.js"></script>
```

Fix F — Human-Readable Sitemap Page

Goal: A styled `/sitemap.html` page listing all pages, for SEO and navigation.

New file: `sitemap.html`

- Sections: Main Pages, Locations (fetched from API), Gallery Albums (from API), Documents (from API)
- Matches IFSA design (dark-mode, Tailwind, amber accents)
- Add "Sitemap" link to footer of every page

Fix G — Dark/Light Mode on All Pages

Goal: Every page remembers and applies the theme — no flash of wrong theme on load.

New file: `theme.js` — place in project root:

```
js
```

```

(function () {
  const KEY = 'ifsa_theme';
  const root = document.documentElement;
  const saved = localStorage.getItem(KEY) || 'dark';
  root.classList.toggle('dark', saved === 'dark');
  root.classList.toggle('light', saved === 'light');

  window.toggleTheme = function () {
    const next = root.classList.contains('dark') ? 'light' : 'dark';
    root.classList.toggle('dark', next === 'dark');
    root.classList.toggle('light', next === 'light');
    localStorage.setItem(KEY, next);
    document.querySelectorAll('.theme-icon')
      .forEach(el => { el.textContent = next === 'dark' ? '🌑' : '🌒'; });
  };

  window.addEventListener('storage', e => {
    if (e.key === KEY && e.newValue) {
      root.classList.toggle('dark', e.newValue === 'dark');
      root.classList.toggle('light', e.newValue === 'light');
    }
  });
})();

```

Every HTML page: Add as the FIRST script in `<head>`:

```

html

<script src="/theme.js"></script>

```

Toggle button HTML (add to every page's navbar):

```

html

<button onclick="toggleTheme()" class="theme-toggle-btn"
  aria-label="Toggle dark/light mode" title="Toggle theme">
  <span class="theme-icon">🌑</span>
</button>

```

Files to change: `theme.js` (new), all `.html` pages

11. Future Roadmap

Features planned for future phases, ordered by impact.

Priority	Feature	Impact	Effort
 Critical	Deploy HTTPS via nginx	Everything breaks without it	Medium
 Very High	Razorpay online fee payment	Closes the "interested → paying" loop	Medium
 Very High	Student login portal	Transforms site from brochure to utility	Hard
 High	Enquiry form + email notification	Captures leads who won't use WhatsApp	Easy
 High	WhatsApp booking confirmation	Auto-confirms trial bookings	Easy
 High	Google Reviews badge	Live rating from verified source	Easy
 High	Grading syllabus page	SEO win + student resource	Medium
 Medium	Export bookings button	Admin downloads Excel from UI	Easy
 Medium	Booking filter/search	Filter by status/branch/date	Easy
 Medium	Admin stats dashboard card	Quick view of monthly enquiries	Easy
 Medium	Rich text announcements (Quill.js)	Bold, links, line breaks in notices	Easy
 Low-Med	Self-hosted Inter font	Remove Google Fonts dependency	Easy
 Low	Image blur placeholders	Smooth fade-in for gallery	Easy
 Low-Med	Video thumbnail generation (ffmpeg)	Gallery shows video previews	Medium
 Medium	JSON-LD on About + Pricing pages	More Google rich results	Easy
 Low	Keyboard-accessible mobile menu	WCAG 2.1 AA compliance	Easy
 High	Rate limiting on booking endpoint	Prevent spam submissions	Easy
 High	Blog / News section	Long-tail SEO, shareable content	Hard
 Medium	Achievements dedicated page	Credibility + parent trust	Easy
 Medium	Multi-admin roles	Scale to more branches/staff	Hard

12. Troubleshooting

Problem	What to Check / Do
Admin panel locked out	Wait 15 minutes (5 failed attempts triggers lockout)
Session expired	Log back in at <code>/admin.html</code>
Photo uploaded but not showing	Hard refresh the page (Ctrl+Shift+R / Cmd+Shift+R)
Website shows "Loading..." forever	Server (Node.js) may be down — run <code>pm2 status</code> or <code>node server.js</code> on the server
Student can't find a document	Check Section 3 in admin — confirm the file is listed there
Deleted something by mistake	No undo — re-upload from your phone/computer
Google Maps not showing for a branch	Re-copy the embed URL from Google Maps → Share → Embed a map
API returns 401 Unauthorized	JWT token expired — log out and log back in
API returns 404	Wrong endpoint URL, or the item doesn't exist
File upload fails	Check file size (photos: auto-compressed; videos: max 100MB; PDFs: no limit set)
PWA not installing on iOS	HTTPS must be active — the service worker won't register over HTTP
Theme flashes wrong colour on load	<code>theme.js</code> must be the FIRST script in <code><head></code>
WhatsApp messages not sending	Check <code>wa-config.json</code> for correct gateway URL + API key

Restarting the Server

```
bash
```



```
# If using PM2 (recommended for production)
pm2 restart ifsa
pm2 logs ifsa --lines 50    # view recent logs

# If running manually
pkill -f "node server.js"
node server.js &
```

Changing the Admin Password

Run this on the server (not in the browser):

```
bash

node change-password.js
```

Follow the prompts. This updates the hashed password in the environment.

13. Security Checklist

Run through this list before going live and after any major change.

- ☐ HTTPS is active via nginx (mandatory — PWA won't work without it)
 - ☐ Admin password is strong (12+ characters, letters + numbers + symbols)
 - ☐ Admin panel URL is not linked anywhere on the public website
 - ☐ Never share the admin password over WhatsApp or email in plain text
 - ☐ Log out of admin panel when done, especially on shared computers
 - ☐ Rate limiting is active: 5 login attempts per 15 minutes per IP
 - ☐ JWT tokens expire after 8 hours (sessionStorage cleared on tab close)
 - ☐ All file uploads are validated by MIME type (not just extension)
 - ☐ Path traversal protection is active (`path.basename` on all filenames)
 - ☐ HTTP security headers are set: `X-Content-Type-Options`, `X-Frame-Options`, `Referrer-Policy`
 - ☐ `robots.txt` blocks `/admin.html` and `/uploads/` from crawlers
 - ☐ Activity log is checked regularly for unexpected actions
 - ☐ Booking endpoint has rate limiting (pending — add 3 submissions/phone/day limit)
-

This guide covers the full IFSA website as of May 2026.

For code changes, provide this document to your developer — all bugs and planned features are

fully specified above.